

Goodness-of-fit Tests

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The Problem

- A **goodness-of-fit test** measures how well the observed data match a fitted (assumed) model.
- We compare the **observed** values to the **expected** (predicted) values under the model.
- General hypotheses:

$$H_0 : \text{Model } M_0 \text{ fits the data,} \quad H_A : \text{Model } M_0 \text{ does not fit.}$$

- This is a **nonparametric** test: it does not require the population to be normal.

Idea: Classification into Categories

- Each observation in a sample is classified into one of a finite number of K **categories**.
- Let p_i be the probability (or proportion) that an observation falls in category i , $i = 1, \dots, K$.

Illustration: blood type

Each observation on blood type falls into one of 4 categories: O, A, B, AB. We may test, for instance,

$$H_0 : p_O = 0.45, p_A = 0.35, p_B = 0.15, p_{AB} = 0.05.$$

- The test statistic measures how different observed counts are from expected counts under H_0 ; a chi-squared distribution gives the p -value $\Rightarrow$ **chi-squared goodness-of-fit test**.

Three Types of Problems

- (1) Discrete variable, the p_i are **completely specified** by H_0 (no parameters to estimate).
- (2) Discrete variable, the p_i depend on **unknown parameters** θ (e.g. Poisson family with μ , Binomial family with p) $\Rightarrow$ estimate θ from the sample.
- (3) **Continuous** variable (e.g. Normal family) $\Rightarrow$ partition the range into K intervals to reduce to the discrete case.

All three use the same test statistic

$$Q^2 = \sum_{\text{all cells}} \frac{(\text{observed} - \text{expected})^2}{\text{expected}},$$

differing only in how $E_i = np_i$ is computed and in the degrees of freedom.

Binomial and Multinomial Experiments

- **Binomial experiment:** each trial results in success (S) or failure (F).
- **Multinomial experiment:** each trial results in one of K possible outcomes ($K \geq 2$).

Notation

- N_i : observed count in category i (observed cell frequencies).
- $E_i = np_i$: expected count in category i under H_0 (expected cell frequencies).
- Large discrepancies between N_i and E_i indicate lack of fit.

Hypotheses and Test Statistic

$H_0 : p_1 = p_{10}, p_2 = p_{20}, \dots, p_K = p_{K0} \quad H_1 : \text{at least one } p_i \neq p_{i0}.$

Test statistic

$$Q^2 = \sum_{i=1}^K \frac{(N_i - np_{i0})^2}{np_{i0}}.$$

Theorem (Pearson)

If H_0 is true and n is large, Q^2 converges in distribution to a chi-squared random variable with $K - 1$ degrees of freedom:

$$Q^2 \xrightarrow{d} \chi_{(K-1)}^2.$$

Condition: $E_i = np_{i0} \geq 5$ for all i (if violated, pool categories with small expected counts).

Decision Rule

- The test is always **upper-tailed**: a larger Q^2 is stronger evidence against H_0 .
- Reject H_0 at significance level α when

$$Q_{obs}^2 > \chi_{\alpha, (K-1)}^2.$$

- Or use the p -value:

$$p = \mathbb{P}\left(\chi_{(K-1)}^2 \geq Q_{obs}^2\right), \quad \text{Reject } H_0 \text{ when } p \leq \alpha.$$

Test procedure

- 1 Draw a sample of size n , define K categories $\Rightarrow$ observed frequencies O_j .
- 2 State H_0 : $p_j = p_{j0}$; compute expected frequencies $E_j = np_{j0}$ (check $E_j \geq 5$).
- 3 Compute $Q^2 = \sum (O_j - E_j)^2 / E_j$.
- 4 Reject H_0 if $Q^2 > \chi_{\alpha, K-1}^2$ (or p -value $\leq \alpha$).

Example 1: Mendelian Genetics Ratios

Problem

Under Mendel's laws, the four phenotypes should occur in the ratios

$$H_0 : p_1 = \frac{9}{16}, p_2 = \frac{3}{16}, p_3 = \frac{3}{16}, p_4 = \frac{1}{16}.$$

Observed on $n = 1611$ plants: $n_1 = 926$, $n_2 = 288$, $n_3 = 293$, $n_4 = 104$. At $\alpha = 5\%$, are the data consistent with Mendel's laws?

Expected counts: $E_i = np_{i0} = 906.2, 302.1, 302.1, 100.7$.

$$Q_{obs}^2 = \sum_{i=1}^4 \frac{(n_i - E_i)^2}{E_i} = 1.473.$$

Example 1: Conclusion and R Check

- $\chi^2_{0.05,3} = 7.815$. Since $Q^2_{obs} = 1.473 < 7.815 \Rightarrow$ **fail to reject H_0** .
- $p\text{-value} = \mathbb{P}(\chi^2_3 \geq 1.473) = 0.688 > 0.05$.
- Conclusion: the data are **quite consistent** with Mendel's laws.

Check in R

```
obs    <- c(926, 288, 293, 104)
probs  <- c(9, 3, 3, 1)/16
chisq.test(x = obs, p = probs)
```

```
# X-squared = 1.4687, df = 3, p-value = 0.6895
```

Example 2: Testing Whether a Die is Fair

Problem

A die is rolled 60 times. We want to test whether the die is fair.

Face i	1	2	3	4	5	6
O_i (observed)	8	12	11	9	10	10
E_i (expected)	10	10	10	10	10	10

$$H_0 : p_1 = \cdots = p_6 = \frac{1}{6} \text{ (fair die)}, \quad Q_{obs}^2 = \sum_{i=1}^6 \frac{(O_i - 10)^2}{10} = 1.0.$$

Since $\chi_{0.05,5}^2 = 11.07$ and $Q_{obs}^2 = 1.0 < 11.07 \Rightarrow$ fail to reject H_0 : no significant evidence the die is unfair.

```
obs <- c(8,12,11,9,10,10); probs <- rep(1/6,6)
chisq.test(x = obs, p = probs)    # X-sq=1, df=5, p=0.9626
```

Example 3: Blood Types in a Population

Problem

The blood types of 500 randomly selected people from a region:

	A	B	AB	O
Count	75	150	15	260

According to a medical reference, the population proportions of blood types are 0.18, 0.28, 0.05, 0.49. At $\alpha = 1\%$, are the blood types in this population consistent with the medical reference?

$$H_0 : p_A = 0.18, p_B = 0.28, p_{AB} = 0.05, p_O = 0.49.$$

Example 3: Solution

Expected counts $E_i = np_{i0}$:

	A	B	AB	O
O_i	75	150	15	260
E_i	90	140	25	245

All $E_i \geq 5$, so the condition is satisfied.

$$Q_{obs}^2 = \frac{(75 - 90)^2}{90} + \frac{(150 - 140)^2}{140} + \frac{(15 - 25)^2}{25} + \frac{(260 - 245)^2}{245} = 8.13.$$

Degrees of freedom = $K - 1 = 3$; critical value $\chi_{0.01,3}^2 = 11.345$.

Since $Q_{obs}^2 = 8.13 < 11.345 \Rightarrow$ **fail to reject** H_0 at the 1% level: the blood types in the sampled population are **consistent** with the medical reference.

```
obs <- c(75,150,15,260); probs <- c(0.18,0.28,0.05,0.49)
chisq.test(x = obs, p = probs)
```

Example 4: Illness and Season of Birth (Uniform Test)

Problem

100 psychiatric patients are classified by the season in which they were born:

	Spring	Summer	Autumn	Winter
Count	20	35	20	25

At $\alpha = 1\%$, are patients **uniformly distributed** across the 4 seasons (i.e. is the illness independent of the season of birth)?

$$H_0 : p_{\text{Spring}} = p_{\text{Summer}} = p_{\text{Autumn}} = p_{\text{Winter}} = \frac{1}{4}.$$

Example 4: Solution

$E_i = 100 \times \frac{1}{4} = 25$ for each season.

$$Q_{obs}^2 = \frac{(20 - 25)^2}{25} + \frac{(35 - 25)^2}{25} + \frac{(20 - 25)^2}{25} + \frac{(25 - 25)^2}{25} = 1 + 4 + 1 + 0 = 6.$$

Degrees of freedom = $4 - 1 = 3$; $\chi_{0.01,3}^2 = 11.345$.

Since $Q_{obs}^2 = 6 < 11.345 \Rightarrow$ **fail to reject** H_0 : no evidence that patients are non-uniformly distributed across seasons of birth, at the 1% level.

```
obs <- c(20,35,20,25); probs <- rep(1/4,4)
chisq.test(x = obs, p = probs)
```

Remark

This is a **test for a uniform distribution** – a special case of the GOF test with $p_{i0} = 1/K$ for all i .

When the p_i Depend on Unknown Parameters

- Often H_0 specifies only a **family** of distributions (Poisson, Binomial, Normal, ...) without giving parameter values.
- Example: $H_0 : X \sim \text{Poisson}(\lambda)$ with λ unknown $\Rightarrow$ **estimate** $\hat{\lambda}$ from the sample (usually by maximum likelihood), then compute $\hat{p}_i = \mathbb{P}(X = x_i; \hat{\lambda})$.

Theorem (Fisher)

If m parameters must be estimated from the sample to compute the $\hat{p}_i$, then under H_0 :

$$Q^2 = \sum_{i=1}^K \frac{(O_i - E_i)^2}{E_i} \xrightarrow{d} \chi^2_{(K-1-m)}.$$

Each estimated parameter removes one degree of freedom. The rejection rule is unchanged: $Q^2 > \chi^2_{\alpha, K-1-m}$.

Example 5: Motorcycle Accidents – Poisson Family

Problem

Number of motorcycle accidents per day in District 5 over 80 days:

Accidents/day	0	1	2	3	≥ 4
Number of days	34	25	11	7	3

At $\alpha = 5\%$, does the daily number of accidents follow a Poisson law?

$$H_0 : X \sim \text{Poisson}(\lambda), \quad \lambda \text{ unknown.}$$

$$\text{Estimate: } \hat{\lambda} = \frac{1}{n} \sum O_i x_i = 1.$$

Example 5: Solution

With $\hat{\lambda} = 1$: $p_i = \mathbb{P}(X = x_i) = \frac{e^{-1} 1^{x_i}}{x_i!}$.

x_i	p_i	$E_i = np_i$	O_i
0	0.368	29.44	34
1	0.368	29.44	25
2	0.184	14.72	11
3	0.061	4.88	7
≥ 4	0.019	1.52	3

$$Q_{obs}^2 = \sum_{i=1}^5 \frac{(O_i - E_i)^2}{E_i} = 4.67.$$

Degrees of freedom: $K - 1 - m = 5 - 1 - 1 = 3$ (lose one d.f. for estimating λ).
 $\chi_{0.05,3}^2 = 7.815$.

Since $4.67 < 7.815 \Rightarrow$ fail to reject H_0 : the daily number of accidents is **consistent** with a Poisson law.

Constructing Intervals for a Continuous Variable

- If X is **continuous** with hypothesized density $f_0(x)$, there are no natural categories $\Rightarrow$ **partition the range** into K intervals

$$[a_0, a_1), [a_1, a_2), \dots, [a_{K-1}, a_K).$$

- Probability of falling in cell i under H_0 :

$$p_{i0} = \mathbb{P}(a_{i-1} \leq X < a_i) = \int_{a_{i-1}}^{a_i} f_0(x) dx.$$

- Observed frequency O_i , expected frequency $E_i = np_{i0}$ (require $E_i \geq 5$).
- Choose intervals so that the E_i are **approximately equal** (equiprobable cells).

$$Q^2 = \sum_{i=1}^K \frac{(O_i - E_i)^2}{E_i} \sim \chi_{(K-1-m)}^2,$$

where m is the number of estimated parameters in f_0 ($m = 0$ if the parameters are given).

Example 6: Do Exam Scores Follow a Normal Distribution?

Problem

Exam scores of 200 students. It is claimed that scores follow a normal distribution with mean 75 and standard deviation 8. At $\alpha = 5\%$, test this claim.

Score	(0, 60]	(60, 70]	(70, 80]	(80, 90]	(90, 100]
Students	12	36	90	44	18

$$H_0 : X \sim \mathcal{N}(75, 8^2) \quad (\text{parameters given} \Rightarrow m = 0).$$

Example 6: Computing the Probabilities p_i

With $Z = \frac{X - 75}{8} \sim \mathcal{N}(0, 1)$:

$$p_1 = \mathbb{P}(X \leq 60) = \Phi(-1.875) = 0.0304,$$

$$p_2 = \mathbb{P}(60 < X \leq 70) = \Phi(-0.625) - \Phi(-1.875) = 0.2356,$$

$$p_3 = \mathbb{P}(70 < X \leq 80) = \Phi(1.25) - \Phi(-0.625) = 0.6284,$$

$$p_4 = \mathbb{P}(80 < X \leq 90) = \Phi(2.5) - \Phi(1.25) = 0.0994,$$

$$p_5 = \mathbb{P}(X > 90) = 1 - \sum_{i=1}^4 p_i = 0.0062.$$

Example 6: Computing Q^2 and Conclusion

p_i	$E_i = 200p_i$	O_i
0.0304	6.08	12
0.2356	47.12	36
0.6284	125.68	90
0.0994	19.88	44
0.0062	1.24	18

$$Q_{obs}^2 = \sum_{i=1}^5 \frac{(O_i - E_i)^2}{E_i} = 274.31.$$

Degrees of freedom = $K - 1 - m = 5 - 1 - 0 = 4$; $\chi_{0.05,4}^2 = 9.488$.

Since $274.31 \gg 9.488$ ($p\text{-value} \approx 3.75 \times 10^{-58}$) $\Rightarrow$ **reject** H_0 : the scores do **not** follow $\mathcal{N}(75, 8^2)$.

Example 7: Equiprobable Cells – Data from $\mathcal{N}(50, 100)$

Problem

We have $n = 100$ observations. The sample gives $\hat{\mu} = 50$, $\hat{\sigma} = 10$; we test whether the data are consistent with $\mathcal{N}(50, 100)$ ($m = 2$ estimated parameters), at $\alpha = 5\%$.

Partition into 5 intervals: $[30, 40)$, $[40, 45)$, $[45, 50)$, $[50, 55)$, $[55, 70)$.

Interval	1	2	3	4	5
O_i	8	20	32	25	15
p_i	0.08	0.18	0.28	0.25	0.21
E_i	8.0	18.0	28.0	25.0	21.0

Example 7: Conclusion

$$Q_{obs}^2 = \sum_{i=1}^5 \frac{(O_i - E_i)^2}{E_i} \approx 0 + 0.222 + 0.571 + 0 + 1.714 = 2.507.$$

Degrees of freedom = $K - 1 - m = 5 - 1 - 2 = 2$ (subtract 2 for estimating μ, σ). $\chi_{0.05,2}^2 = 5.99$.

Since $Q_{obs}^2 = 2.507 < 5.99 \Rightarrow$ fail to reject H_0 : the data are **consistent** with $\mathcal{N}(50, 100)$.

Example 8: Exam Completion Times – Equiprobable Cells

Problem

Final exams are supposed to last 120 minutes. The department believes completion time X is normal with $\mu = 100$ minutes, and exactly 90% of students finish within 120 minutes. A sample of $n = 120$ students is recorded.

From the 90th percentile: $\mu + 1.28\sigma = 120 \Rightarrow \hat{\sigma} = \frac{120 - 100}{1.28} \approx 15.63$ (minutes).

Partition into $K = 8$ **equiprobable** intervals with $p_{i0} = 1/8$, so each $E_i = 120 \times \frac{1}{8} = 15$.

Example 8: Data and Computation of Q^2

Interval	1	2	3	4	5	6	7	8
O_i	21	17	12	16	10	15	19	10
E_i	15	15	15	15	15	15	15	15

$$Q_{obs}^2 = \sum_{i=1}^8 \frac{(O_i - E_i)^2}{E_i} = 7.73.$$

Degrees of freedom = $K - 1 - m = 8 - 1 - 2 = 5$ (estimating μ, σ). p -value = $\mathbb{P}(\chi_5^2 \geq 7.73) = 0.1715 > 0.10$.

Fail to reject H_0 : **no evidence** that completion times violate the stated criteria $X \sim \mathcal{N}(100, 15.63^2)$.

```
obs <- c(21,17,12,16,10,15,19,10); n <- sum(obs)
chisq_stat <- sum((obs - n/8)^2/(n/8))
pchisq(chisq_stat, df = 5, lower.tail = FALSE)    # 0.1716
```

Summary of the Three Cases

Case	Compute E_i	Degrees of freedom
(1) p_i fully specified	$E_i = np_{i0}$	$K - 1$
(2) Discrete, estimate m parameters	$E_i = n\hat{p}_i$	$K - 1 - m$
(3) Continuous (K intervals)	$E_i = n\hat{p}_i$	$K - 1 - m$ ($m = 0$ if parameters given)

In every case:

$$Q^2 = \sum_i \frac{(O_i - E_i)^2}{E_i} \sim \chi^2_{(\text{df})} \text{ when } H_0 \text{ is true,} \quad \text{Reject } H_0 \Leftrightarrow Q^2 > \chi^2_{\alpha, \text{df}}.$$

Practical Notes

- Always check $E_i \geq 5$ for every cell; if not, **pool** adjacent categories with small expected counts.
- The GOF test is **upper-tailed only**: an unusually small Q^2 is not strong support for H_0 .
- Each parameter estimated from the *same* data **removes one degree of freedom**.
- For continuous data, prefer **equiprobable** intervals to avoid widely varying E_i .
- In R: `chisq.test(x = observed, p = probs)` handles case (1); with estimated parameters, adjust the d.f. manually via `pchisq(..., df = K-1-m)`.

Summary of Steps

- 1 Identify the variable and split it into K categories (or K intervals if continuous).
- 2 State H_0 : the population follows model M_0 (possibly with unknown parameters θ).
- 3 If needed, estimate $\hat{\theta}$ (m parameters) from the sample; compute $\hat{p}_i$ and $E_i = n\hat{p}_i$.
- 4 Check $E_i \geq 5$; pool cells if necessary.
- 5 Compute $Q_{obs}^2 = \sum (O_i - E_i)^2 / E_i$.
- 6 Compare with $\chi_{\alpha, K-1-m}^2$ (or compute the p -value) and conclude.